Nanotechnology

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To cite this article: Dezhi Wu *et al* 2022 *Nanotechnology* **33** 405205

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Nanotechnology 33 (2022) 405205 (11pp) https://doi.org/10.1088/1361-6528/ac73a4

A flexible tactile sensor that uses polyimide/graphene oxide nanofiber as dielectric membrane for vertical and lateral force detection

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Received 2 March 2022, revised 17 May 2022
Accepted for publication 24 May 2022
Published 14 July 2022

Abstract

Flexible force sensors are of great interest in the fields of healthcare, physiological signals, and aircraft smart skin applications because of their compatibility with curved surfaces. However, the simultaneous detection of multidirectional forces remains an engineering challenge, despite the great progress made in recent years. Herein, we present the development of a flexible capacitive force sensor capable of efficiently distinguishing normal and sliding shear forces. A two-layer electrospun polyimide/graphene oxide (PI/GO) nanofiber membrane is used as the dielectric layer, which is sandwiched between one top electrode and four symmetrically distributed bottom electrodes. This composite membrane has an improved dielectric constant, a reduced friction coefficient, and good compressibility, leading to superior performance that includes high sensitivity over a wide operational range with measured results of 3 MPa^{-1} for 0–242 kPa (0–2.2 N) and 0.92 MPa^{-1} for 242–550 kPa (2.2–5 N) in the normal direction; and better than 1 N^{-1} for 0–3 N in the *x*- and *y*-axis directions. The system also has a low detection limit of 10 Pa, fast response and recovery times of 39 ms and 13 ms, respectively, a good cyclic stability of 10,000 cycles at a pressure of 176 kPa, and promising potential for use in high-temperature environments (200 °C). Moreover, a prototype 4 × 4 sensor array has been fabricated and successfully used in a robotic system to grasp objects and operate a wireless toy car. As such, the proposed system could offer superior capabilities in simultaneous multidirectional force sensing for applications such as intelligent robots, human–machine interaction, and smart skin.

Supplementary material for this article is available online

Keywords: three-dimensional perception, flexible, capacitive force sensor, composite nanofiber membrane

(Some figures may appear in colour only in the online journal)

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1. Introduction

Flexible tactile sensors have broad applications in many fields, such as intelligent robots [1], physiological signal monitoring [2–4], smart skin [5], and human–computer interaction [6–10]. These applications generally demand high sensitivity and fast responses. In previously published works, tactile sensors have shown impressive results [11–13], but there is a critical residual challenge of simultaneously achieving multidirectional tactile detection with high sensitivity and long-term stability [14–16]. In particular, it has been very difficult to progress tactile sensing in different directions under high-pressure conditions [17, 18].

In general, tactile sensors have been based on some key sensing mechanisms: capacitive [19–21], piezoresistive [22–24], piezoelectric [25, 26], optical [27], and triboelectric models [28, 29]. Capacitive sensors generally have the advantages of excellent sensitivity, simple structure, rapid dynamic response, and long-term stability. Meanwhile, a variety of microstructures [30, 31] and nanomaterials [32, 33] have been utilized to improve the performance of capacitive sensors. For example, Bai *et al* [34] proposed an intra-fillable microstructure that reached a high sensitivity of 220 kPa^{-1} over a broad pressure range of 0.08–360 kPa and had an operational temperature limit of 100 °C, which was fabricated using a complex mold transfer process [19]. On the other hand, the electrospinning process provides a simple scheme for the construction of tactile sensors with enhanced sensing performance [35–37]. For example, Lou *et al* [38] demonstrated a sensor using self-assembled 3D films made from Reduced graphene oxide (RGO)-encapsulated P(VDF-TrFe) nanofibers, which had a high sensitivity of 15.6 kPa^{-1} in the range of 30–50 kPa. Jeong *et al* [39] fabricated a sensor based on polydimethylsiloxane (PDMS) and silver nanowires (AgNWs) that had a sensitivity of 16.1 kPa^{-1} over a pressure range of 0–42 kPa. In the field of three-dimensional tactile sensors, Yao *et al* [40] produced a U-shaped groove structure with sensitivities of 1.08 N^{-1} and 1.20 N^{-1} with respect to normal and tangential force ranges of 0.7 N and 0.6 N, respectively. Boutry *et al.* used nanocomposites and nanostructures to optimize the performance of a pressure sensor with pyramid-shaped microstructures [41] to achieve a pressure sensitivity of 0.19 kPa^{-1} at 1 kPa in the normal direction and a pressure sensitivity of 3 kPa^{-1} over the range of 10–20 kPa in the lateral direction. Sun and coworkers [22] proposed a flexible sensor that incorporated Carbon nanotube (CNT)/PDMS nanocomposites and achieved sensitivities of 0.68 kPa^{-1} over a range of 1–5 kPa in the normal direction and 2.3 N^{-1} at 0.6 N for tangential forces. These investigations advanced the sensing performance of three-dimensional tactile sensors, but a larger working range (> 40 kPa) and high sensitivity in both normal and lateral directions are two key limitations.

Here, we propose a flexible three-dimensional capacitive force sensor with one top electrode and four symmetrically distributed bottom electrodes. Two electropsun PI/GO nanofiber membranes are utilized as the dielectric material to allow large deformations in the normal and lateral directions. Changes in the capacitances of the overlap areas between the top and bottom electrodes are used to measure the magnitude and direction of the applied force. High sensitivities were measured, namely 3 MPa^{-1} for the range of 0–242 kPa (0–2.2 N) and 0.92 MPa^{-1} for the range of 242–550 kPa (2.2–5 N) in the normal direction and 1.12 N^{-1} for the range of 0–3 N in the *x*-axis direction. A sensitivity of 1.03 N^{-1} for the range of 0–3 N was measured in the *y*-axis direction. Furthermore, a short response time of 39 ms, a fast recovery time of 13 ms, a low detection limit of 10 Pa, good repeatability of more than 10,000 cycles under 176 kPa, thermostability (200 °C), and excellent recovery characteristics were measured. As seen in figure S1 (available online at stacks.iop.org/NANO/33/405205/mmedia), experimental results show that the double-layer nanofiber dielectric membrane exhibits better performance than those based on either single-layer or triple-layer dielectric structures. As a demonstration example, a 3D force perception sensor based on a 4 × 4 array was constructed for two human–computer interaction applications, namely, grasping objects and the control of a wireless toy car.

2. Experimental details

2.1. Materials

Graphene oxide powder (Aldrich), pyromellitic dianhydride (PMDA) (Aladdin, $M_W = 218.12$ g mol^{-1}), 4,4′-diaminodiphenylether (ODA) (Aladdin, $M_W = 200.24$ g mol^{-1}), PDMS (Dow Corning Sylgard 184), N,N-dimethylacetamide (DMAC) (Sinopharm Chemical Reagent Co., LTD, M_W =87.12 g mol^{-1}), sterile tape (Hons Medical), and copper foil (Wenzhou Chunfeng Electronics, thickness = 10 μm) were purchased and used without modification.

2.2. Preparation of the dielectric layer

The preparation of the 30 wt% polyamic acid (PAA)/GO precursor solution is explained as an example. In the first step, 5.4 mg of GO powder is dissolved in 14.1 g of DMAC solution and put into a water bath in an ultrasonic system (KUDOS, Shanghai) at 53 kHz for 3 h. Subsequently, 3 g of ODA powder is added and magnetically stirred for 1 h and 3.1 g of PMDA powder is added and magnetically stirred at room temperature for 12 h (IKA C-MAG HS 7, Germany).

A commercial four-needle electrospinning machine (E02, Qingzi Nano, China) is used to electrospin the PAA/GO nanofiber membrane. The precursor solution is stored in a syringe connected to four flat needles that each have a diameter of 250 μm. The operating voltage is set to 17 kV and the distance between the needle and the collection roller is 15 cm. The precursor pumping rate is 0.3 ml h^{-1}, and the speed of the collection roller is 108 r min^{-1}. The fabricated PAA/GO nanofiber membrane is placed in a vacuum furnace (VBF-1200X, China) and kept at 100 °C for 1 h (solvent removal), at 250 °C for 1 h (dehydration cyclization), and at 350 °C for 1 h (thermal amination) [42] to convert the PAA/GO nanofiber membrane into PI/GO nanofiber membrane.

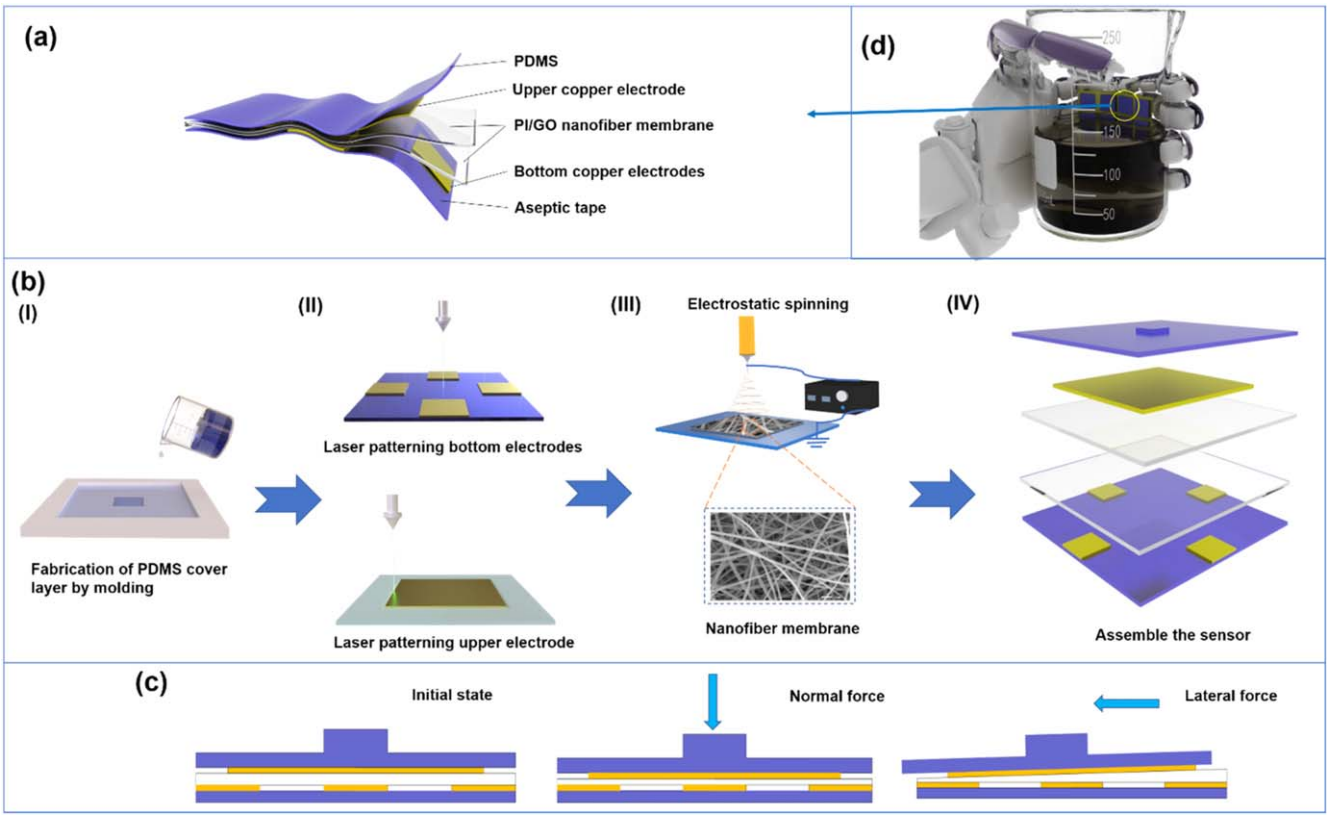

Figure 1. (a) Schematic diagram of the flexible capacitive force sensor, which uses PI/GO nanofiber as a dielectric membrane for vertical and lateral force detection. (b) The fabrication process. (c) Schematic diagram of the change of the electrode plate distance under a normal force and the change of the relative positions and electrode plate distances between the top and bottom electrodes under a lateral force. (d) Demonstration of the application of a prototype sensor array in a robotic manipulator for a grasping operation.

2.3. Fabrication of the three-dimensional flexible pressure sensor

The fabrication process of the tactile sensor is illustrated in figure 1(b). PDMS (Dow Corning Sylgard 184) with a weight ratio of 10:1 by mass is used as the upper substrate of the sensor. First, PDMS is stirred for 1 h by an electric stirrer (IKA C-MAG HS 7, Germany) at a speed of 300 r min^{-1}. Second, oxygen plasma treatment (Alpha Plasma, Germany) is conducted for 10 min. Third, after keeping the PDMS solution at room temperature for 3 h, it is poured into an 8 × 8 mm mold fabricated by a 3D printer (Dimension Elite, USA) and cured in a vacuum oven (DZF-600, Linpinyiqi, Shanghai) at 60 °C for 2 h. Finally, the PDMS layer is peeled off at a thickness of 1 mm.

The copper foil is patterned by a UV laser (JPT-SEAL-355–10W, Shanghai Wavetopsign International Technology Co. Ltd.) at an operating current of 1 A. The four 2 × 2 mm symmetrical bottom electrodes are cut at a speed of 150 mm s^{-1} at 56 kHz, which is then repeated at 50 mm s^{-1} at 80 kHz. The upper electrode is cut into a 6 × 6 mm area by an 80 kHz laser at 150 mm s^{-1}. The upper electrode and PDMS are then closely attached, and the 100 μm-thick double-layer nanofiber membrane is placed between the upper and bottom electrodes. Finally, the sensor is packaged with PDMS to complete the fabrication process of the flexible three-dimensional tactile sensor.

2.4. Fabrication of the 4 × 4 sensor array

The 4 × 4 sensor array (figure 5(a)) has one upper (60 μm, single-layer) and one lower (120 μm, double-layer) gold-plated electrode Flexible printed circuit (FPC) (Shenzhen Deep Circuit Board Technology Co. Ltd.) (see figure S4). The upper FPC is closely attached to the PDMS, the bottom FPC is glued to the sterile tape, and the 100 μm-thick double-layer nanofiber membrane is placed between the upper and lower FPCs. Finally, the 4 × 4 sensor array is packaged using PDMS to complete the process.

2.5. Characterization and measurements

The morphologies of the nanofiber membrane are characterized using scanning electron microscopy (SEM Su70, Germany). A digital dynamometer (HP-100, Aidebao, China) with a range of 0–100N is used to measure the applied force. A three-axis moving stage (PT-SD203, Paidiwei, China) is used to apply the lateral force. The dynamic force is provided by an electromechanical universal test machine (E43.104, MTS SYTEMS, China) with a 0–5 N force sensor, and the corresponding capacitance change is measured by a precision L (Inductance), C (Capacitance), R (Resistance) meter (Agilent E4980AL, Keysight Technologies, USA) at a sampling rate of 500 kHz. To ensure the accuracy and reliability of the test, three sensors are fixed to the sample platform of the

electromechanical universal test machine and connected to the LCR meter. A computer is used to record capacitance changes. The friction coefficient between the copper foil and the nanofiber membrane is calculated using a home-made slope platform (figure S5) based on the transformation relationship between gravitational potential energy and thermal energy.

3. Results and discussion

3.1. Structure and sensing principle of the sensor

Figure 1(a) shows a prototype flexible capacitive pressure sensor consisting of a PDMS cover layer (8 × 8 mm), a copper-plate top electrode (6 × 6 mm), a double-layer electrospun PI/GO (0.0885 wt%) nanofiber membrane, four symmetrically distributed copper-plate bottom electrodes (2 × 2 mm), and the sterile tape as the base substrate. Figure 1(b) illustrates the key fabrication process steps and several important fabrication factors which were investigated to achieve the best nanofiber characteristics, including the electrospinning voltage of 17 kV, the needle-to-collector distance of 15 cm, the polymer solution pumping rate of 0.3 ml h^{-1}, and the dielectric membrane thickness of 100 μm. When a normal force is applied, the change of capacitance is determined by the distance between the two electrode plates and the change of the dielectric constant of the membrane material. When a lateral force is applied, the relative positions of the top and bottom electrodes change, resulting in a change in the capacitance, while the gap distance between the two electrode plates and the dielectric constant of the membrane material may also change due to the induced torque, changing the capacitance as shown in figure 1(c).

3.2. Three-dimensional sensing performance

To evaluate the performance of the prepared sensor, it is necessary to measure and calculate the response of the capacitive sensor. The sensitivity in the normal direction (S_N), the sensitivity in the X-axis direction (S_X), and the sensitivity in the Y-axis direction (S_Y) under various normal pressures P and lateral forces F can be expressed as follows:

$$S_N = \frac{\Delta C_1/C_1 + \Delta C_2/\Delta C_2 + \Delta C_3/C_3 + \Delta C_4/C_4}{4\Delta P} \quad (1)$$

$$S_X = \frac{\Delta C_1/C_1 - \Delta C_4/C_4}{\Delta F} \quad (2)$$

$$S_Y = \frac{\Delta C_2/C_2 - \Delta C_3/C_3}{\Delta F}, \quad (3)$$

where $C_1 \sim C_4$ stand for the initial values of the four capacitances, respectively, and $\Delta C_1 \sim \Delta C_4$ represent the changes of the four capacitances.

As shown in figure 2(a), when pressure is applied at 25 °C and 200 °C (PDMS is stable at 200 °C), the sensitivity in the normal direction of the sensor is 3 MPa^{-1} in the pressure range of 0–242 kPa (0–2.2 N) and 0.92 MPa^{-1} in the pressure range of 242–550 kPa (2.2–5 N). The performance of the sensor is stable at 200 °C and produces very similar results to those at 25 °C, showing the possible broad application prospects of the sensor at a range of temperatures. Figure 2(b) shows that the sensitivity of a prototype sensor subjected to a shear force in the x-axis direction is 1.12 N^{-1} in the range of 0–3 N. It can be seen that the capacitances of the the capacitors in the left and right positions (C_1 and C_4) can change linearly (increase or decrease) with respect to the applied force when a shear force is applied in the x-axis direction toward the C_1 location of the sensor. On the other hand, both the top and bottom (C_2 and C_3) capacitors have small changes in capacitance. This phenomenon is expected, as the left and right electrodes experience changes in their relative positions, producing greater (or smaller) overlap areas for C_1 (C_4) and no changes in the overlap areas for C_2 and C_3 when a lateral force is applied is in the x-axis direction toward the C_1 location of the sensor. Similar results are observed if a shear force is applied in the y-axis direction, which yields a sensitivity of 1.03 N^{-1} (figure 2(c)). These results show that the sensor can sense the magnitude and direction of the applied force with good accuracy in both the normal and lateral directions. Compared with sensors that use single-layer nanofiber membranes, sensors that use double-layer nanofiber membranes have an extra layer of air in the dielectric layer, which results in a lower equivalent compressive modulus of the dielectric layer [43] and increased normal and lateral sensitivities. It can be seen that as the number of dielectric layers increases from two to three, the frictional drag of the nanofiber membranes increases, thereby reducing the lateral sensitivity of sensors assembled using triple-layer nanofiber membranes (figure S1). Therefore, the double-layer nanofiber membrane is better than the other types.

We now consider tactile sensing characteristics. Human tactile receptors can sense signals in less than 70 ms [44]. Here, the dynamic response under a rapid external load of 220 kPa is shown in figure 2(d); the results show a response time of 39 ms and a recovery time of 13 ms. The detection limit of the sensor is characterized by reducing the applied normal force and observing the capacitance changes. It is found that when a petal (10 mg) is placed on the sensor, the capacitance change can still be clearly observed (as shown in figure 2(e)) in the form of a small detected pressure variation of 10 Pa. Figures 2(f), (g) further show that the sensor has good responses under stepped loading and unloading conditions, which were applied using loading intervals of 55 kPa. The sensor maintained good repeatability after 10 000 cycles of external loading at 176 kPa. Compared with the previous studies displayed in table 1, the proposed sensor has a wide working range, high lateral sensitivity, and great potential for use in high-temperature environments.

The influences of GO additives on the deformation, compression modulus, and friction coefficient of the nanofiber membrane were investigated. The sensitivity of the sensor with respect to the applied normal pressure is given by:

$$S = \frac{\Delta C / C_0}{\Delta P} = \frac{\left(\frac{\varepsilon_r' \varepsilon_{air} A}{d'} - \frac{\varepsilon_r \varepsilon_{air} A}{d}\right) \Big/ \frac{\varepsilon_r \varepsilon_{air} A}{d}}{\Delta P} = \frac{\frac{\varepsilon_r'}{\varepsilon_r} \cdot \frac{d}{d'} - 1}{\Delta P}, \quad (4)$$

where ε_r' and d' refer to the dielectric constant and the gap between the plates under the applied pressure, respectively,

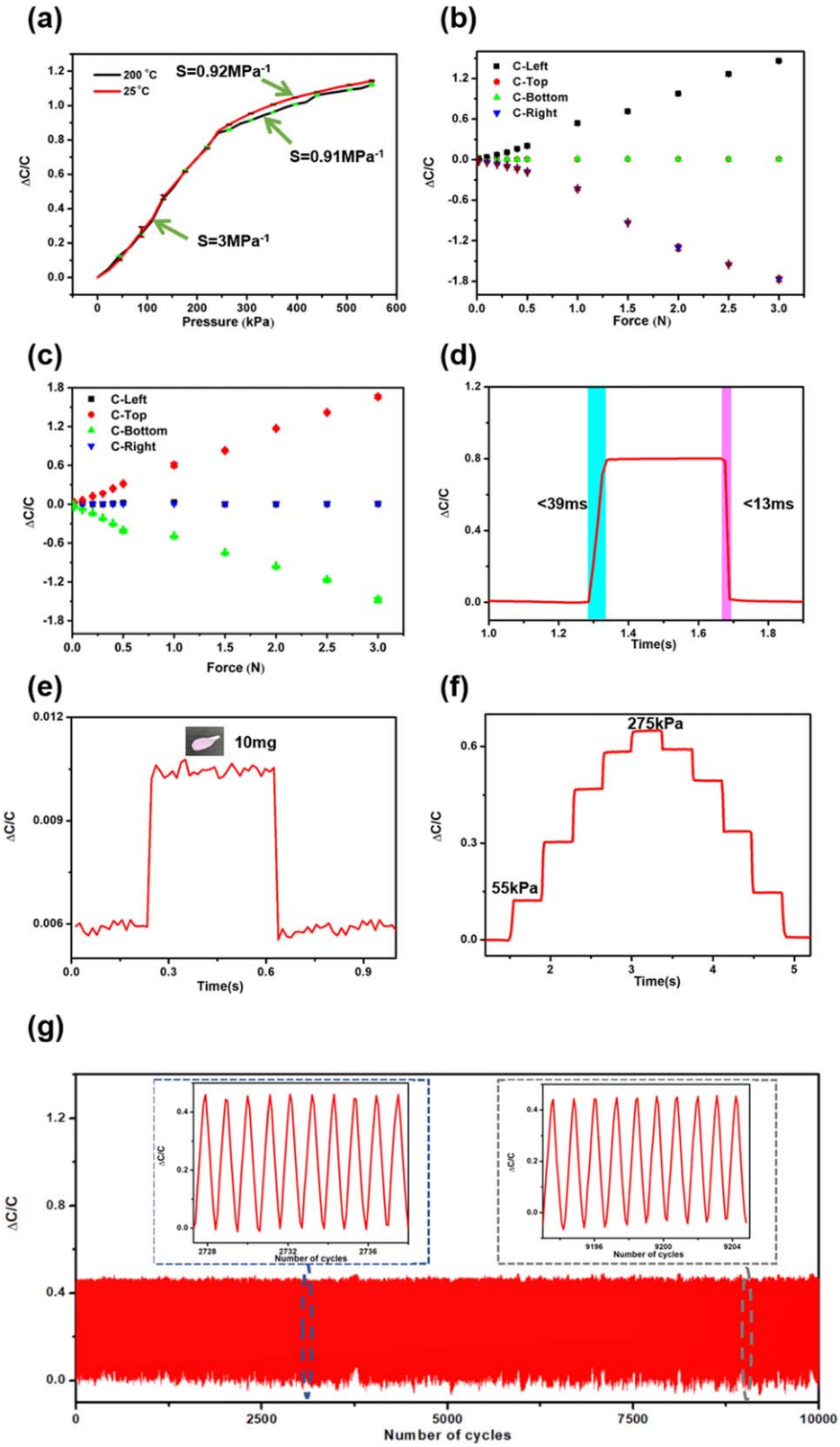

Figure 2. Typical sensing characteristics. (a) Measured capacitance changes of a prototype sensor at operational temperatures of 25 °C and 200 °C with respect to applied pressures of 0–550 kPa. (b) Measured capacitance changes of the four capacitors of the sensor with respect to applied forces of 0–3 N when the prototype sensor is subjected to a shear force in the *x*-axis direction. (c) Measured capacitance changes of the four capacitors of the sensor with respect to applied forces of 0–3 N when the prototype sensor is subjected to a shear force in the *y*-axis direction. (d) The capacitance response and recovery time of a prototype sensor under a rapid loading of 220 kPa. (e) The capacitance change of the sensor for a 10 Pa normal pressure. (f) The capacitance changes under stepped loading and unloading pressures at 55 kPa intervals. (g) The capacitance change curve of a prototype sensor after 10 000 loading cycles at 176 kPa of normal pressure.

Table 1. A comparison between the performance of the sensor reported in this article and previously reported triaxial force sensors, including normal sensitivity (S_N), lateral sensitivity (S_L), response time (T_1), recovery time (T_2), working range (W_R), and working temperature (W_T).

Author	S_N	S_L	T_1	T_2	W_R	W_T
Wu *et al* (this work)	3 MPa^{-1} (0–242 kPa) 0.92 MPa^{-1} (242–550 kPa)	1.12 N^{-1} (0–3 N)	39 ms	13 ms	0–550 kPa	~200 °C
Ji *et al* [45]	0.0124 kPa^{-1} (0–12 kPa)	0.0752 N^{-1} (0.015–0.5 N)	100 ms	100 ms	0–600 kPa	Room temperature
Viry *et al* [17]	0.91 kPa^{-1} (0–0.5 kPa)	0.34 N^{-1}	40 ms		0–190 kPa	Room temperature
Yao *et al* [40]	1.08 N^{-1} (0–0.7 N)	1.2 N^{-1} (0–0.6 N)	40 ms	100 ms	0–0.7 N	Room temperature
Lee *et al* [46]	2.9% mN^{-1}	3% mN^{-1}			0–131 kPa	Room temperature
Sun *et al* [22]	0.68 kPa^{-1} (1–5 kPa)	2.3 N^{-1} (0.05–0.6 N)	3.1 ms	6.8 ms	0–6 kPa	Room temperature
Chen *et al* [47]	6.33 kPa^{-1} (0.1–3kPa)	1.28 N^{-1} (0–0.9N)	59 ms	22 ms	0.1–18.8 kPa	Room temperature

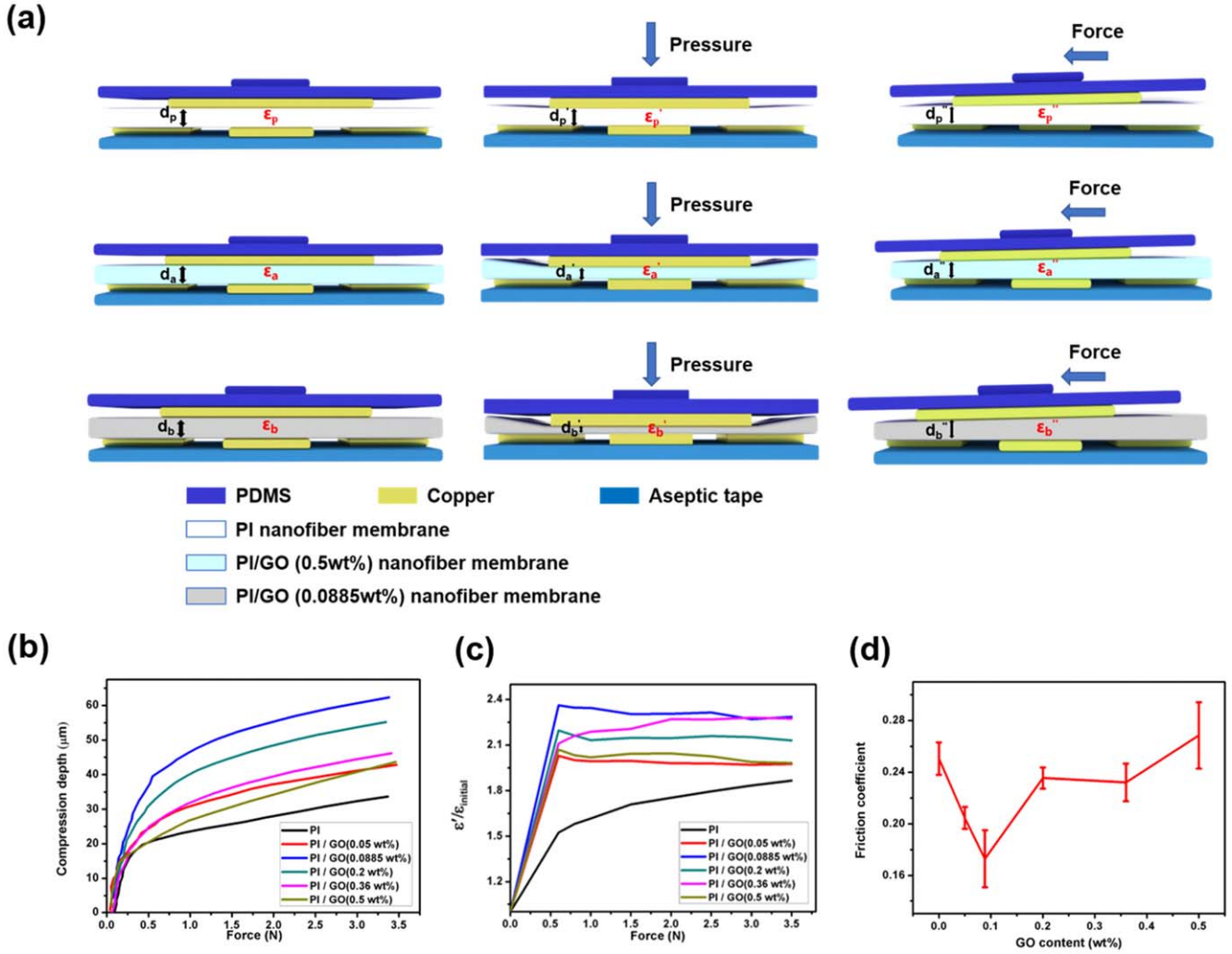

Figure 3. (a) Schematic diagram of the changes of the electrode plate spacing, electrode-facing area, and effective dielectric constant when sensors with different graphene oxide contents are subjected to normal force and shear force. (b) Measured applied normal force and compressed depth of a 110 μm-thick double-layer nanofiber membrane for different graphene oxide contents. (c) The relative effective permittivity versus applied normal force of double-layer nanofiber dielectric membranes that have different graphene oxide contents. (d) Friction coefficient of the copper foil and the PI/GO nanofiber membrane for different graphene oxide contents.

and ΔP represents the change of pressure applied to the sensor. When normal pressure is applied, the membrane is compressed and the gaps between the plates of the four capacitors decrease to d'. At the same time, the relative dielectric coefficient ε_r may increase to ε_r', since the nanofibers are squeezed in the process and air cavities are reduced, as shown in figure 3(a). On the other hand, under a lateral force applied in the x-axis direction toward C_1, torque is generated at the PDMS boundary that reduces (increases) the gap of C_1 (C_4) due to the tilting of the top electrode, while the area of overlap (A) of C_1 (C_4) increases (decreases) due to the movement of the top electrode. At the same time, the relative dielectric coefficient of C_1 (C_4) increases (decreases).

Figure 3(b) shows that the addition of GO initially results in a reduction of the compression modulus of the nanofiber membrane, which continues until the GO content is 0.0885 wt%. Subsequently, the compression modulus of the nanofiber membrane increases, but its value is still less than that of pure PI as the GO content increases. This is the result of nanoparticles which introduce free volumes between polymer chains such that the polymer chains can easily approach each other, which results in a low compressive modulus. As the concentration of the composite material increases, the nanoparticles begin to agglomerate inside the nanofibers, increasing the compressive modulus [48]. As such, an adequate amount of GO is utilized to reduce the compression modulus of the nanofiber membrane to a level that produces large gap changes under an applied pressure, resulting in improved sensitivity in the capacitance measurements, as predicted by equation (4). Furthermore, the large gap changes of the dielectric layer also result in a smaller volume fraction of air, which increases the dielectric constant of the membrane and improves the sensivity in the capacitance measurements, as predicted by equation (4).

The relationship between gap deformation and applied force follows the fundamental physics of a parallel capacitor. Prototype double-layer 100 μm-thick nanofiber membranes fabricated with different graphene oxide contents are tested

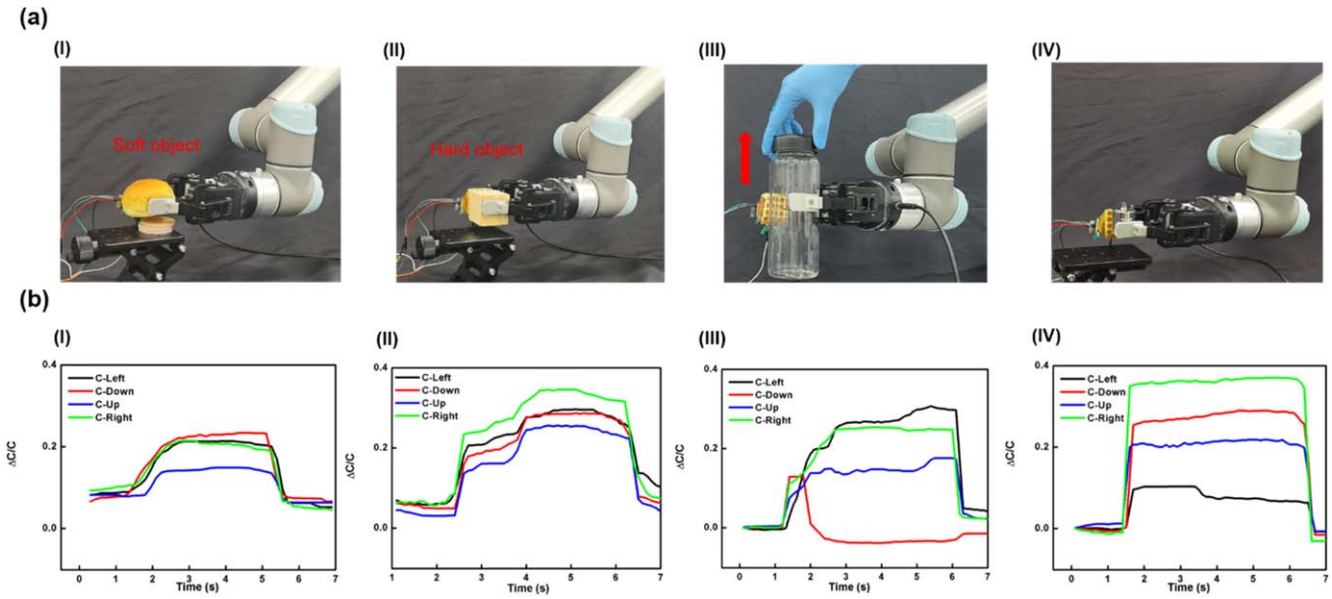

Figure 4. Real-time monitoring of manipulator grasping. (a) Optical images of a manipulator grasping different objects. (b) The electrical signals detected by the sensor when the manipulator grasps (I) a bread roll, (II) a cube, (III) a water bottle, (IV) a weight.

under different magnitudes of applied normal force and the effective dielectric constants of the membranes with respect to force are derived, as shown in figure 3(c). It can be seen that as the applied force increases, the effective dielectric constant of the nanofiber membranes increases as the air in the dielectric layer is compressed. It can be observed that when the GO content is 0.0885 wt%, the effective dielectric constant of the dielectric membrane is maximized. Analytically, GO can form a three-dimensional network structure, and the conductivity and dielectric constant of the composite material increases when the concentration is near the percolation threshold. If the GO content exceeds the percolation threshold, the filler agglomerates, leading to a rapid decrease in the dielectric constant. Therefore, the addition of an appropriate graphene oxide content can increase the effective dielectric constant of the membrane and the sensitivity of the sensor.

The addition of GO can reduce the friction coefficient of the copper foil–nanofiber film interface, as shown in figure 3(d). As the GO content increases, the friction coefficient gradually decreases and reaches a minimum when the GO content is 0.0885 wt%. This is attributed to the layered structure of graphene oxide and the low van der Waals force between layers, which allow intrinsic super-lubricity to form a certain amount of transfer film on the polymer surface [49, 50], thereby reducing the friction coefficient. The small friction coefficient helps the movements of the electrode to achieve high sensitivity when subjected to lateral forces.

The flexible force sensor can effectively monitor the process by which a robotic manipulator grasps different objects. When the manipulator first touches the object, the normal force gradually increases, and the values of the four capacitors of the sensor gradually increase. Subsequently, the manipulator grasps the object with minimum force, creating a constant capacitance reading. As the manipulator releases the object, the capacitance decreases and returns to its initial state. Figure 4(a) shows optical images of a robotic manipulator and different objects. Figures 4(b)(I $\sim$ II) illustrate processes in which the robotic manipulator grasps a soft bread roll (supplementary video S1) and a hard cube (supplementary video S2). It can be seen that the robotic manipulator responds slowly to the soft object, causing small relative capacitance changes (small forces), but it responds fast to hard objects, causing large relative capacitance changes (large forces). The capacitances of the four capacitors show different changes due to the fabrication variations of the sensor and the variations in the grasping location of the manipulator. If the object is heavy or if the object has slippery surfaces and the holding force used by the manipulator is not high enough, a possible 'slipping' process may occur during this type of operation, as shown in figures 4(b)(III $\sim$ IV).

To validate the responses of the sensor to arbitrary triaxial forces, we apply 0–1 N arbitrary triaxial forces to the sensor and get the information relating to the forces, angles, and corresponding capacitances. The computer converts the corresponding capacitance information into decoupled Fx, Fy, and Fz. By comparing the original force curve and the decoupled force curve (figure S6), we can see that the small decoupling error confirms the excellent ability of the sensor to perceive arbitrary triaxial forces.

As illustrated in figure 4(b)(IV), when the manipulator steadily holds a weight and a slipping force is suddenly applied to the weight, the capacitance of one of the capacitors drops quickly as a result and then remains stable, demonstrating its ability to monitor slippage. We also apply an increasing lateral force to a water bottle grabbed by the manipulator to simulate an invasive grabbing process (supplementary video S3), and it can be seen that the capacitance gradually decreases and then the total change of capacitance

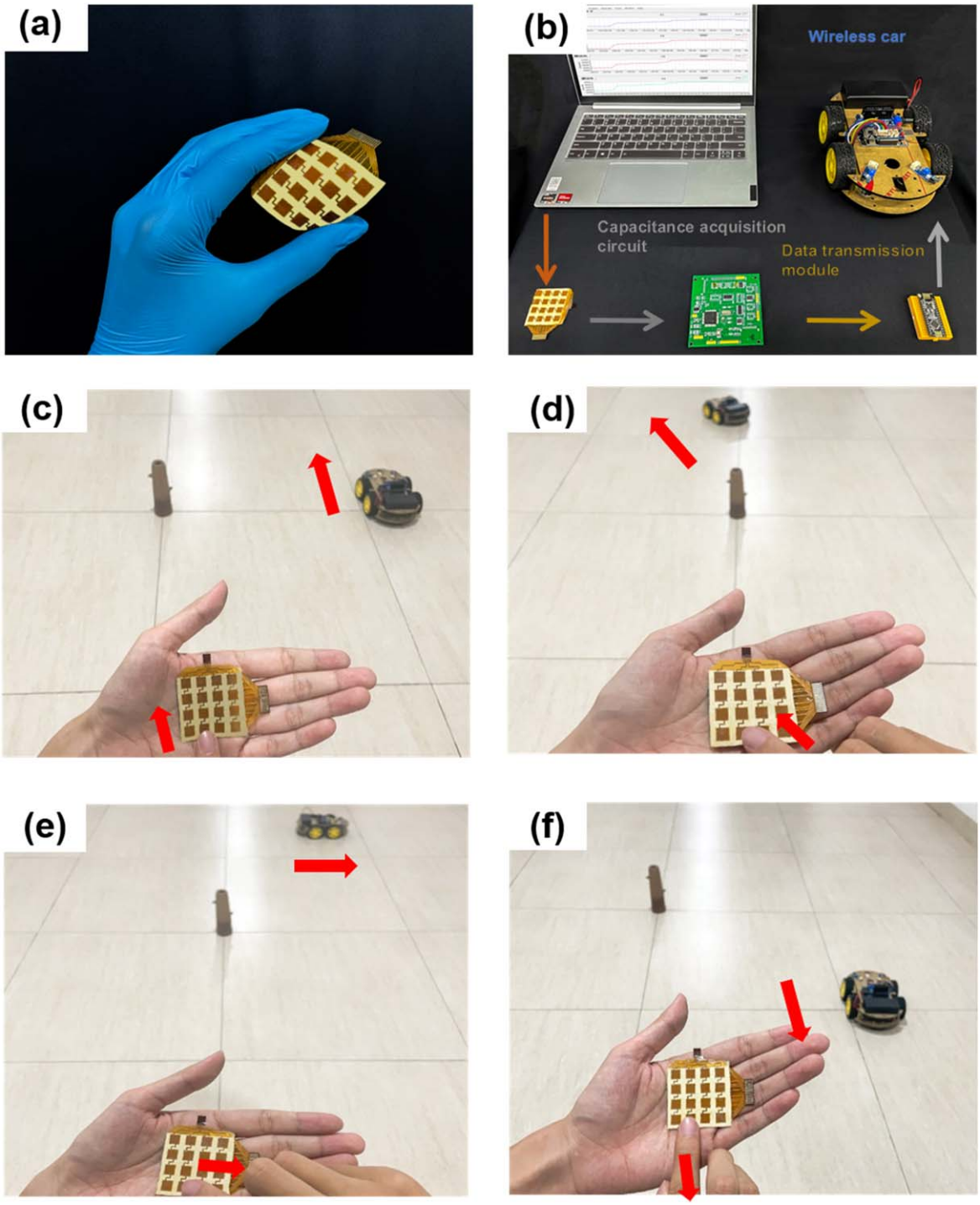

Figure 5. The PI/GO-based three-dimensional flexible pressure sensor is applied to control a toy car's movement. (a) The 4 × 4 sensor array. (b) Flexible control sensor system, including the capacitance acquisition circuit, a data transmission module, a wireless car, and a computer. (c~f) The wireless car moves in the direction indicated by the flexible pressure sensor through wireless communications.

becomes negative, indicating that the effect of the lateral force on the capacitor is greater than that of the normal force.

The capacitive tactile sensor array is also used as a flexible pad to control the movements of a wireless toy car (figure 5(a)). The control system of the wireless toy car consists of the capacitance acquisition circuit, a data transmission module, a wireless toy car, and a computer (figure 5(b)). The capacitance acquisition circuit measures the variation of each capacitor, and the data transmission module processes the signals from the sensors and determines the direction of movement of the wireless car. As depicted in figures 5(c)–(f) and supplementary video S4, the wireless car moves under control of the flexible sensor array, demonstrating the development of flexible sensors for human–computer interaction and intelligent robot applications.

4. Conclusions

In summary, a flexible capacitive tactile sensor with good stability and performance over a large working range exhibits good sensing characteristics after 10 000 cycles of 176 kPa loading. Owing to the excellent characteristics of its PI/GO nanofiber membranes, the sensor can produce measurement results over a large working range, achieving fast response and recovery times, a low detection limit, and good thermo-stability. Experimental results demonstrate that GO can significantly reduce the compressive modulus, effective dielectric constant, and friction coefficient of the nanofiber dielectric layer, which greatly enhances the sensing performance. The demonstration examples show that the sensor can help a robotic manipulator to stably grasp different objects and to control the direction of movement of a toy car wirelessly via a 4 × 4 flexible sensor array. As such, this work could potentially assist promising applications in the fields of intelligent robots and human–computer interactions.

Acknowledgments

This work is supported in part by the National Natural Science Foundation of China (52075464), the Laboratory Open Fund of Beijing Smart-chip Microelectronics Technology Co. Ltd. and Program Granted by Special Funds from the Central Government to Guide Local Scientific and Technological Development (2021Szvup068).

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Conflicts of interest

There are no conflicts to declare.

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